#4

a) Raise (T_1)'s RHS to $1+\varepsilon$, all three rows. Tight system:

$y_1+y_2=1+\varepsilon$, $y_2+y_3=1$, $y_1+y_3=1$.

From the last two, $y_2=1-y_3$ and $y_1=1-y_3$; substituting into the first gives $2-2y_3=1+\varepsilon$, so

$$y_3=\frac{1-\varepsilon}{2}=0.5-\frac{\varepsilon}{2}, \qquad y_1=y_2=0.5+\frac{\varepsilon}{2}$$

Cost:

$$z(1+\varepsilon)=5\left(0.5+\frac{\varepsilon}{2}\right)+5\left(0.5+\frac{\varepsilon}{2}\right)+6\left(0.5-\frac{\varepsilon}{2}\right)=$$

$$=(2.5+2.5+3)+(2.5+2.5-3)\varepsilon=8+2\varepsilon$$

Slope $dz/d\varepsilon=2$, exactly $\pi_1=2$ from our task file - the dual price of (T_1) is the local rate of change of z with it's own right-hand side, verified directly.

b) Validity range. We need $0\le y_1, y_2\le 1$ and $0\le y_3\le 1$, i.e $0\le 0.5+\varepsilon/2\le 1$ and $0\le 0.5-\varepsilon/2\le 1$, both giving $-1\le\varepsilon\le 1$. So the linear formula $z=8+2\varepsilon$, and hence $\pi_1=2$, is valid for $-1\le\varepsilon\le 1$ (at $\varepsilon=\pm 1$ one of the variables hits its bound, $y_3=0$ or $y_1=y_2=1$, and the slope could change beyond that point).

c) Meaning of π_1. If (π_1) means "the first customer must be covered", $\pi_1 = 2$ says: tightening that coverage requirement by one unit (e.g requiring it be covered twice over, or relaxing it) changes the team's optimal total cost by exactly 2 per unit of requirement - it is the marginal cost of how strictly VM / customer coverage for that pair is enforced

Danial Kabdulla

IT-2408